

Calculating the Inverse of a 3×3 Matrix

Let A be the matrix. The formula for its inverse A^{-1} and for an auxiliary matrix $\text{adj}(A)$ called the adjoint of A (or in some books the adjugate of A) are

$$A^{-1} = \frac{1}{|A|} \text{adj}(A) = \frac{1}{|A|} \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix}^T$$

In the formula, A_{ij} is the cofactor of the element a_{ij} in the matrix, i.e., its minor with its sign changed by the checkerboard rule.

Steps to calculate A^{-1} :

1. Calculate the matrix of minors.
2. Change the signs of the entries according to the checkerboard rule.
3. Transpose the resulting matrix. This gives $\text{adj}(A)$.
4. Divide every entry by $|A|$ (provided that $|A| \neq 0$).

Example 2.2

Find the inverse of

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Solution:

First compute the cofactors matrix:

$$\begin{pmatrix} 1 & 1 & -1 \\ 0 & 2 & 0 \\ 1 & -1 & 2 \end{pmatrix}$$

Then the adjugate matrix $\text{adj}(A)$ is

$$\text{adj}(A) = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & -1 \\ -1 & 0 & 1 \end{pmatrix}$$

Hence the inverse is

$$A^{-1} = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 1 & -\frac{1}{2} \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$$

To check your result, verify that

$$AA^{-1} = I$$

or equivalently

$$A \cdot \text{adj}(A) = |A|I$$

Inverse of a 2×2 Matrix

For a general matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

its inverse is

$$A^{-1} = \frac{1}{|A|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

where $|A| = ad - bc$.

a)

$$A = \begin{pmatrix} 1 & 0 \\ 3 & 2 \end{pmatrix}$$

Determinant:

$$|A| = (1)(2) - (0)(3) = 2$$

Adjugate matrix:

$$\text{adj}(A) = \begin{pmatrix} 2 & 0 \\ -3 & 1 \end{pmatrix}$$

Inverse:

$$A^{-1} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{3}{2} & \frac{1}{2} \end{pmatrix}$$

b)

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & -1 & 1 \\ 1 & 3 & 2 \end{pmatrix}$$

Cofactor matrix:

$$\begin{pmatrix} -5 & -3 & 7 \\ 2 & 0 & -1 \\ 4 & 3 & -5 \end{pmatrix}$$

Adjugate matrix:

$$\text{adj}(A) = \begin{pmatrix} -5 & 2 & 4 \\ -3 & 0 & 3 \\ 7 & -1 & -5 \end{pmatrix}$$

Inverse:

$$A^{-1} = \frac{1}{3} \begin{pmatrix} -5 & 2 & 4 \\ -3 & 0 & 3 \\ 7 & -1 & -5 \end{pmatrix}$$

Proof of the formula for the inverse matrix

To show:

$$AA^{-1} = I$$

or equivalently

$$A \operatorname{adj}(A) = |A|I$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{pmatrix} = \begin{pmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{pmatrix}$$

To prove (iv), it will be enough to look at two in the top row. According to the rule of multiplying the two matrices on the left, what we have to show is that

$$a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} = |A| \quad (15)$$

$$a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23} = 0 \quad (16)$$

Matrix Inverses

These two equations are both evaluations of determinants by Laplace expansions. The first equation (15) evaluates the determinant on the left below by cofactors of the first row; the second equation (16) evaluates the determinant on the right below by cofactors of the second row. (Notice that the cofactors of the determinant do not care what is actually in the second row, since to calculate them you only need to know the other two rows.)

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \quad \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

The two equations (15) and (16) now follow, since the determinant on the left is just $|A|$, while the determinant on the right is 0, since two of its rows are the same.

The procedure we have given for calculating an inverse works for $n \times n$ matrices, but gets to be too cumbersome if $n \geq 3$, and other methods are used. The calculation of A^{-1} for reasonably sized A is a standard package in computer algebra programs and matrices.

Unfortunately, social scientists often want the inverse of very large matrices, and for this special techniques have had to be devised, which produce approximate but acceptable results.

Session 12: Equations of Planes

Equation of Planes

Recall an equation of the form

$$ax + by + cz = d$$

defines a plane.

1) Plane through the origin with normal vector

$$\vec{N} = \langle 1, 5, 10 \rangle$$

If P is in the plane,

$$\iff \vec{OP} \cdot \vec{N} = 0$$

$$\iff 2x + 5y + 10z = 0$$

Plane through $P_0(2, 1, -1)$ with normal vector

$$\vec{N} = \langle 1, 5, 10 \rangle$$

A point $P(x, y, z)$ is in the plane if:

$$\vec{P_0P} \cdot \vec{N} = 0$$

$$\langle x - 2, y - 1, z + 1 \rangle \cdot \langle 1, 5, 10 \rangle = 0$$

$$(x - 2) + 5(y - 1) + 10(z + 1) = 0$$

$$x + 5y + 10z = -3$$

Vectors and Planes

Is $\vec{v}$ perpendicular to $\vec{N}$?

$$\vec{v} \cdot \vec{N} = \langle 1, 2, -1 \rangle \cdot \langle 1, 1, 3 \rangle = 1 + 2 - 3 = 0$$

Yes, so $\vec{v}$ is parallel to the plane.

Check if $(1, 2, -1)$ is in the plane

$$x + y + 3z = 5$$

Substitute:

$$1 + 2 + 3(-1) = 0 \neq 5$$

So the point is not in the plane.

3×3 Linear System Example

$$\begin{cases} 9x + z = 1 \\ x + y = 2 \\ x + 2y + 3z = 3 \end{cases}$$

The first two planes intersect in a line. The third plane intersects that line in a point (if the system has a solution).

Matrix Form

A linear system can be written as

$$AX = B$$

If A is invertible, the solution is

$$X = A^{-1}B$$